

MSTAR PRACTICE WORKSHEET**Lesson 7-6: Determine Distance on the Coordinate Plane**

Grade 6 Mathematics · Standard 6.NOS.9

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 **SELECTED RESPONSE — CORRECT: C**What is the distance between $(-4, 3)$ and $(2, 3)$?

- ☐ A 2 units
- ☐ B 4 units
- ☒ C 6 units
- ☐ D 8 units

Both points have $y = 3$, so this is a horizontal distance. $|2 - (-4)| = |2 + 4| = 6$ units.

Question 2 **SELECTED RESPONSE — CORRECT: C**What is the distance between $(5, -2)$ and $(5, 4)$?

- ☐ A 2 units
- ☐ B 4 units
- ☒ C 6 units
- ☐ D 10 units

Both points have $x = 5$, so this is a vertical distance. $|4 - (-2)| = |4 + 2| = 6$ units.

Question 3 **SELECTED RESPONSE — CORRECT: C**

A rectangle has vertices at (1, 1), (5, 1), (5, 4), (1, 4). What is its perimeter?

- ☐ A 8 units
- ☐ B 12 units
- ☒ C 14 units
- ☐ D 20 units

Width = $5 - 1 = 4$. Height = $4 - 1 = 3$. Perimeter = $2(4 + 3) = 14$.

Question 4 **SELECTED RESPONSE — CORRECT: D**

Two points are at (-3, -7) and (-3, 5). What is the distance between them?

- ☐ A 2 units
- ☐ B 7 units
- ☐ C 8 units
- ☒ D 12 units

Same $x = -3$. Distance = $|5 - (-7)| = |5 + 7| = 12$ units.

Question 5 **SELECTED RESPONSE — CORRECT: A**

What is the distance between (2, 3) and (2, 8)?

- ☒ A 5 units
- ☐ B 5.5 units
- ☐ C 6 units
- ☐ D 11 units

The x -coordinates match, so distance = $|8 - 3| = 5$ units.

Question 6

SELECTED RESPONSE — CORRECT: C

What is the distance between $(-4, 1)$ and $(3, 1)$?

- ☐ A -7 units
- ☐ B 1 unit
- ☒ C 7 units
- ☐ D 12 units

The y-coordinates match, so distance = $|3 - (-4)| = 7$ units.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7

TWO-PART QUESTION

Part A — correct answer: D

A chest sits at $(-3, 2)$ and a cave at $(4, 2)$. How far apart are they?

- ☐ A 1 unit
- ☐ B 2 units
- ☐ C 5 units
- ☒ D 7 units

Both points share the same y-value, so the distance is horizontal: $|-3 - 4| = 7$ units.

- **A:** That subtracts the digits without regard to sign. Count from negative 3 up to zero, then on to 4.
- **B:** That is the shared y-value, not a distance. Measure along x.
- **C:** Both points sit at $y = 2$, so the gap is purely horizontal. Count along x from negative 3 to 4.

Part B — correct answer: A

Why can this distance be found by subtraction alone?

- ☒ A The two points share the same y-coordinate, so the segment between them is horizontal.
- ☐ B Subtraction works for any two points on the plane.
- ☐ C Both coordinates are whole numbers.
- ☐ D One of the points is in Quadrant II.

When two points share a coordinate, the segment runs straight along a grid line and its length is the difference of the coordinates that DO change. A slanted segment needs more than subtraction.

Question 8

WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Subtracting a Negative

Priya found the distance between $(-3, 2)$ and $(4, 2)$. She wrote: 'Just subtract: $4 - 3 = 1$, so they are 1 unit apart.'

Explain Priya's misconception and give the correct distance.

Model answer: $|4 - (-3)| = |4 + 3| = 7$ units. Priya treated -3 as 3. Counting on the number line from -3 to 0 is 3 units and 0 to 4 is 4 more, confirming 7.

2 points	Student identifies that Priya dropped the negative sign on -3 , sets up $ 4 - (-3) = 4 + 3 = 7$, and confirms by counting grid units from -3 to 4.
1 point	Student gives 7 units without explaining the dropped negative sign.
0 points	Student agrees with Priya or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9

WRITTEN RESPONSE · 2 POINTS

A rectangle has vertices at $(-4, -2)$, $(3, -2)$, $(3, 5)$, and $(-4, 5)$. Find the length, width, perimeter, and area of the rectangle. Show your work using absolute value.

Model answer: The width comes from the two x-values, $|3 - (-4)| = 7$, and the height comes from the two y-values, $|5 - (-2)| = 7$, so this rectangle is actually a square. That gives a perimeter of $2(7 + 7) = 28$ units and an area of $7 \times 7 = 49$ square units.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.